Camera-Based Handwashing Assessment

Timed speaker script

Group 43 | 13-minute target | English narration

Timing plan

Slide narration: 8 minutes, including the short introductions before and after playback. Recorded demonstration video: 5 minutes. Total target: 13 minutes. Rehearse once to align the spoken delivery with the timing cues.

Slide	Topic	Video time
1	Camera-Based Handwashing Assessment	00:00-00:25
2	What the system assesses	00:25-01:05
3	The self-recorded dataset	01:05-01:55
4	Why we model temporal boundaries	01:55-02:30
5	YOLO + Adaptive Boundary-Aware GRU	02:30-03:30
6	Adaptive action confirmation	03:30-04:20
7	Recorded system demonstration	04:20-09:20
8	Recognition and boundary results	09:20-10:25
9	Complete action chains	10:25-11:10
10	The local deployment choice	11:10-11:55
11	Conclusions and next evaluation	11:55-13:00

Recording setup

Use Presenter View to follow the slide notes. Have the prepared demonstration video ready for playback. Check that the screen recording captures its picture and any existing audio.

At 04:20, play the prepared five-minute demonstration video. At 09:20, return to slide 8. The video is not yet embedded in this PPT because its file has not been provided.

SLIDE 01 | 00:00-00:25 | 25 seconds

Camera-Based Handwashing Assessment

Hello, we are Group forty-three. Our project uses a local camera to assess six-step handwashing. It recognizes hand gestures, reconstructs their order, and measures recognized washing time. We will explain the model, demonstrate the application, and discuss the results and deployment trade-offs.

Presenter cue: Start the presentation timer. Introduce the team without reading every name.

SLIDE 02 | 00:25-01:05 | 40 seconds

What the system assesses

A correct gesture prediction is only one part of the task. A person can perform recognizable actions while skipping a step, changing the order, or stopping too early. Our application therefore evaluates the complete procedure. It recognizes six actions and a background class, builds the observed action sequence, and checks total recognized washing time. Completion requires all six steps in the standard order and at least forty seconds of recognized actions. Background time is excluded. This is an assessment of visible procedure, rather than a direct measurement of cleanliness.

Presenter cue: Point to the action examples, then to the order and duration requirements.

SLIDE 03 | 01:05-01:55 | 50 seconds

The self-recorded dataset

Our target-scene dataset contains six thousand nine hundred and eighty-seven sampled frames from thirty-two recordings. Sampling at about five frames per second captures motion variation and transitions. The recordings include standard sequences, reordered or missing actions, and isolated examples of steps four and five. We split by video, so neighboring frames stay in the same partition. Fourteen videos support training, eight support validation, and ten form the reported test set. We distinguish definite action labels from uncertain transition frames. Those uncertain frames remain explicit in the annotations instead of becoming background labels by default.

Presenter cue: Emphasize frame annotation and the split by video. Avoid describing adjacent frames as independent samples.

SLIDE 04 | 01:55-02:30 | 35 seconds

Why we model temporal boundaries

We first studied the public PSKUS dataset to check feasibility. On that preliminary comparison, macro F1 near transitions increased from eighty-three point three for YOLO to ninety point one with GRU, and ninety-one point zero with early boundary awareness. This supported using temporal context and transition supervision. We then trained the final systems on our self-recorded data. The final model set contains no PSKUS-trained weights. The public-data experiment explains the design choice, while our own scene supplies the final evaluation.

Presenter cue: The plot shows preliminary transition-region macro F1, which differs from the final output-boundary F1 metric.

SLIDE 05 | 02:30-03:30 | 60 seconds

YOLO + Adaptive Boundary-Aware GRU

Here is the complete architecture. YOLO extracts visual features from each sampled frame. A unidirectional GRU carries information forward through the sequence. Two heads share this temporal representation. The classification head estimates the current action, while the boundary head estimates where an action changes. Boundary-guided decoding produces candidate segments, and adaptive confirmation decides which segments belong in the final chain. This separates frame recognition from confirmation of an action instance. The recurrent state also supports incremental processing during a camera session. We compare this complete method with standalone YOLO, ordinary YOLO plus GRU, and VideoMAE. Each baseline answers a different question about visual recognition, temporal context, or video-window modeling.

Presenter cue: Trace the diagram from the frame thumbnail through the two heads to the final chain. Explain the flow without reading layer dimensions.

SLIDE 06 | 03:30-04:20 | 50 seconds

Adaptive action confirmation

Brief prediction changes can create extra actions even when most frames are correct. We smooth labels over about one second, then group adjacent labels into segments. Adaptive confirmation compares each segment's duration and confidence with the recording's own statistics. An unusually short and uncertain segment can remain unconfirmed. This replaces the fixed three-second rule. With fewer than four comparable segments, duration rejection is skipped. The example removes an extra step four while preserving the observed order. We will now play a five-minute recording of the application in use.

Presenter cue: Finish the introduction before 04:20, then start the prepared demonstration video.

04:20-09:20 | VIDEO PLAYBACK

Play the prepared five-minute demonstration video. Resume the narration below when it ends.

SLIDE 08 | 09:20-10:25 | 65 seconds

Recognition and boundary results

That was our system demonstration. We now turn to the quantitative results. The quantitative comparison uses six hundred and thirty-two aligned valid time centers from ten videos. Accuracy and macro F1 use the same reference labels for every model. Boundary F1 measures predicted changes using one-to-one matching with transition tolerance. Our method reaches ninety-six point five two percent accuracy, ninety-three point seven one macro F1, and ninety-four point four boundary F1. Relative to ordinary GRU, the largest improvement is boundary F1, which rises by six point eight one percentage points. VideoMAE achieves the highest values on all three measures. We report that clearly. These are complete-system results under their actual resolutions and training protocols, so they do not establish a controlled architecture-only advantage.

Presenter cue: Read the proposed row and the boundary improvement. Do not read every table cell.

SLIDE 09 | 10:25-11:10 | 45 seconds

Complete action chains

Frame scores do not fully describe sequence quality. YOLO matches only two of the ten complete reference chains, ordinary GRU matches six, and both our final pipeline and VideoMAE match all ten. The examples show that a correct prediction can describe an incorrect procedure. Our model preserves a missing step and a reordered sequence instead of repairing them into the standard order. Only our pipeline uses adaptive confirmation on the dense five-frame-per-second stream. These results therefore compare complete methods, including their decoding, rather than an isolated change to the neural network.

Presenter cue: Point to the four bars, then distinguish prediction correctness from procedure correctness.

SLIDE 10 | 11:10-11:55 | 45 seconds

The local deployment choice

We retain our model for the local camera application because deployment also depends on size and temporal input requirements. It has about ten point six four million parameters, compared with eighty-six point two three million for VideoMAE. Its recurrent path consumes a frame and an existing state. VideoMAE uses a sixteen-frame window, requiring about three seconds of initial context at our sampling rate. Centered predictions also use future frames. VideoMAE can run locally and stream delayed predictions, but its workflow differs from immediate incremental feedback. The parameter comparison supports our choice without claiming a measured speed or energy advantage.

Presenter cue: Present VideoMAE as the accuracy reference. Do not claim it is impossible to deploy or that parameter ratio equals speed ratio.

SLIDE 11 | 11:55-13:00 | 65 seconds

Conclusions and next evaluation

The project delivers a working camera application, annotated data, four comparison models, and reproducible evaluation records. Our selected pipeline combines visual recognition with learned boundaries and adaptive action confirmation. It achieves all ten reference chains in the reviewed evaluation and supports local feedback on both order and recognized duration. The scope of that evidence matters. We have inspected these recordings repeatedly, including correcting labels after reviewing predictions. The results therefore describe internal validation in the recorded scene. The next evaluation should freeze labels and settings, then use new people, camera positions, and lighting. It should also measure sustained throughput on the same hardware. Our contribution is a complete, inspectable assessment system with a clear deployment rationale. Thank you.

Presenter cue: Finish at 13:00. Keep the final slide visible as the recording ends.